



HORMONES, SEXUAL HEALTH & MISC

PNC-27

PNC-27 – 5 mg or 10 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	PNC-27; chimeric peptide comprising a p53-derived segment fused to the penetratin membrane-residency sequence
SYNONYMS / ALIASES	PNC-27; p53-penetratin chimeric peptide
CAS NUMBER	1159861-00-3
MOLECULAR FORMULA	C188H293N53O44S
MOLECULAR WEIGHT	Approximately 4032 g/mol
AMINO ACID SEQUENCE	PPLSQETFSDLWKLLKKWKMRRNQFWVKVQRG (32 residues)
SOURCE	Synthetic (solid-phase peptide synthesis)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=98% by HPLC (confirm against batch COA)
CONTAINER	5 mg or 10 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, use within approximately 28 days
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. Protect reconstituted solution from light.

STRUCTURAL NOTES

PNC-27 is a 32-residue chimeric construct. The N-terminal segment corresponds to a region of the p53 transactivation domain; the C-terminal segment is the penetratin sequence derived from the Antennapedia homeodomain, included as a membrane-residency element. The peptide is strongly basic and contains a single methionine, which is susceptible to oxidation; store cold and protect from air.

SELECTED PEER-REVIEWED REFERENCES

1. Sarafraz-Yazdi E. et al. — *PNC-27, a Chimeric p53-Penetratin Peptide Binds to HDM-2 in a p53 Peptide-like Structure, Induces Selective Membrane-Pore Formation and Leads to Cancer Cell Lysis* — *Biomedicines*, 2022;10(5):945 — DOI: 10.3390/biomedicines10050945

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.